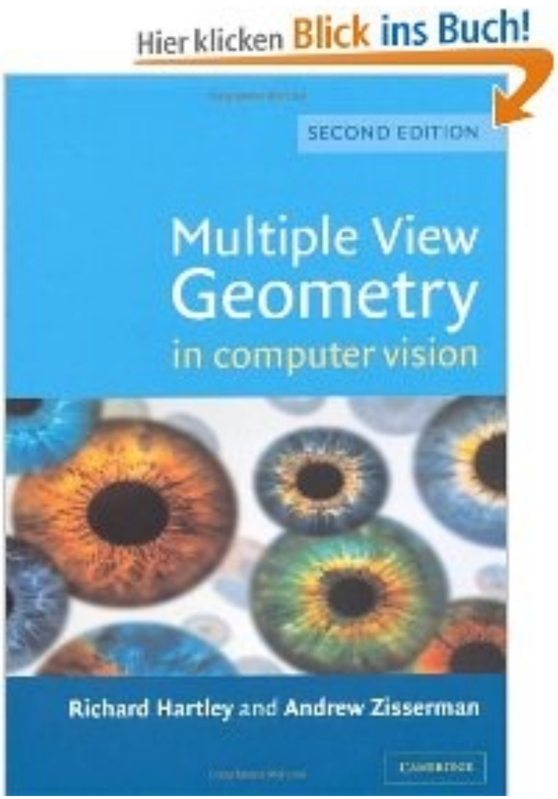
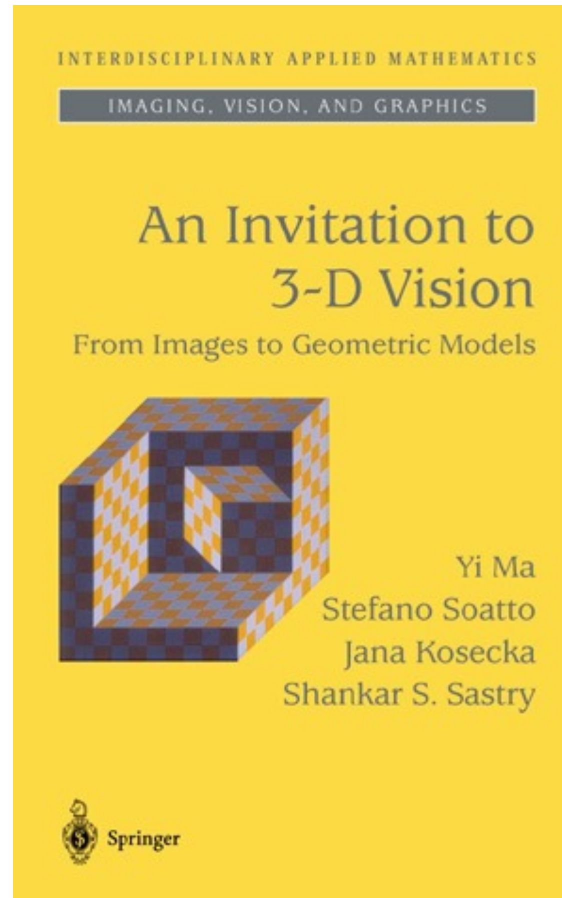
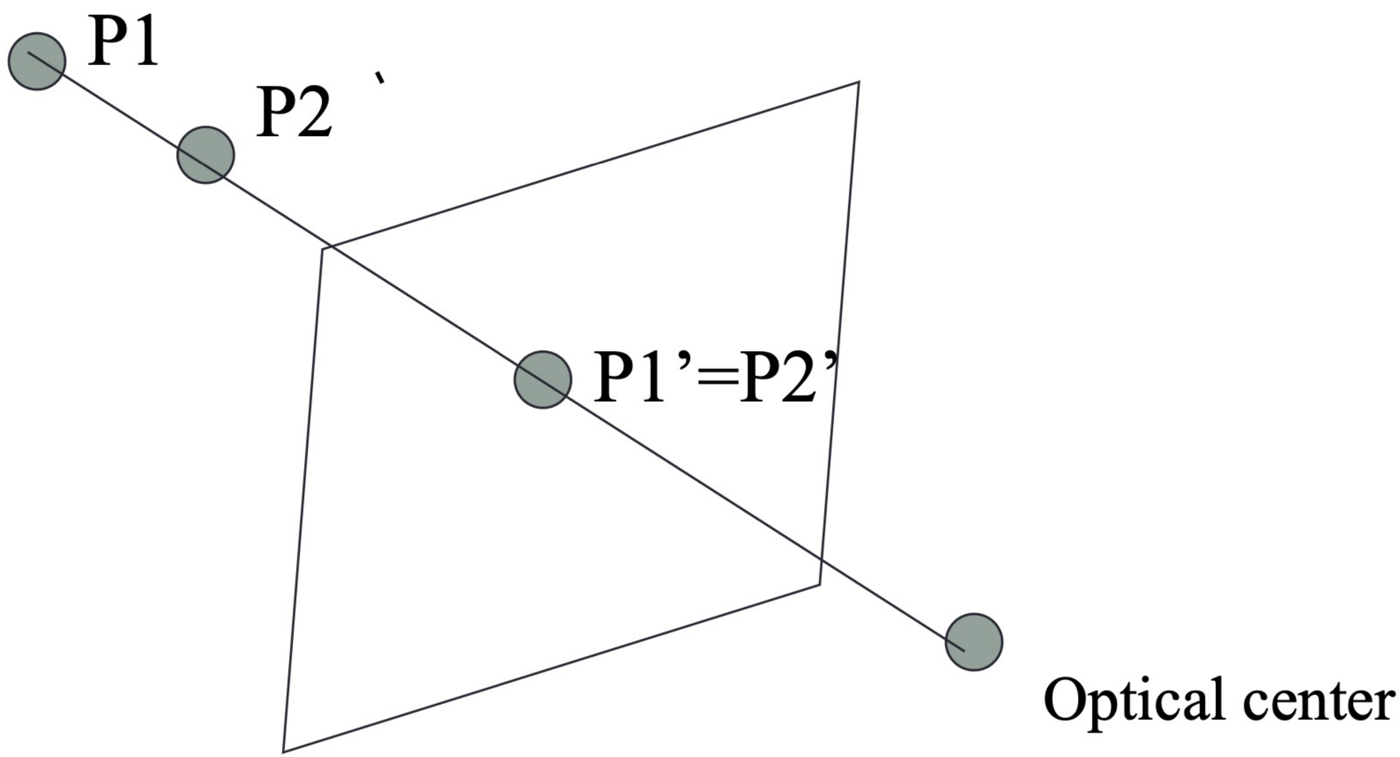
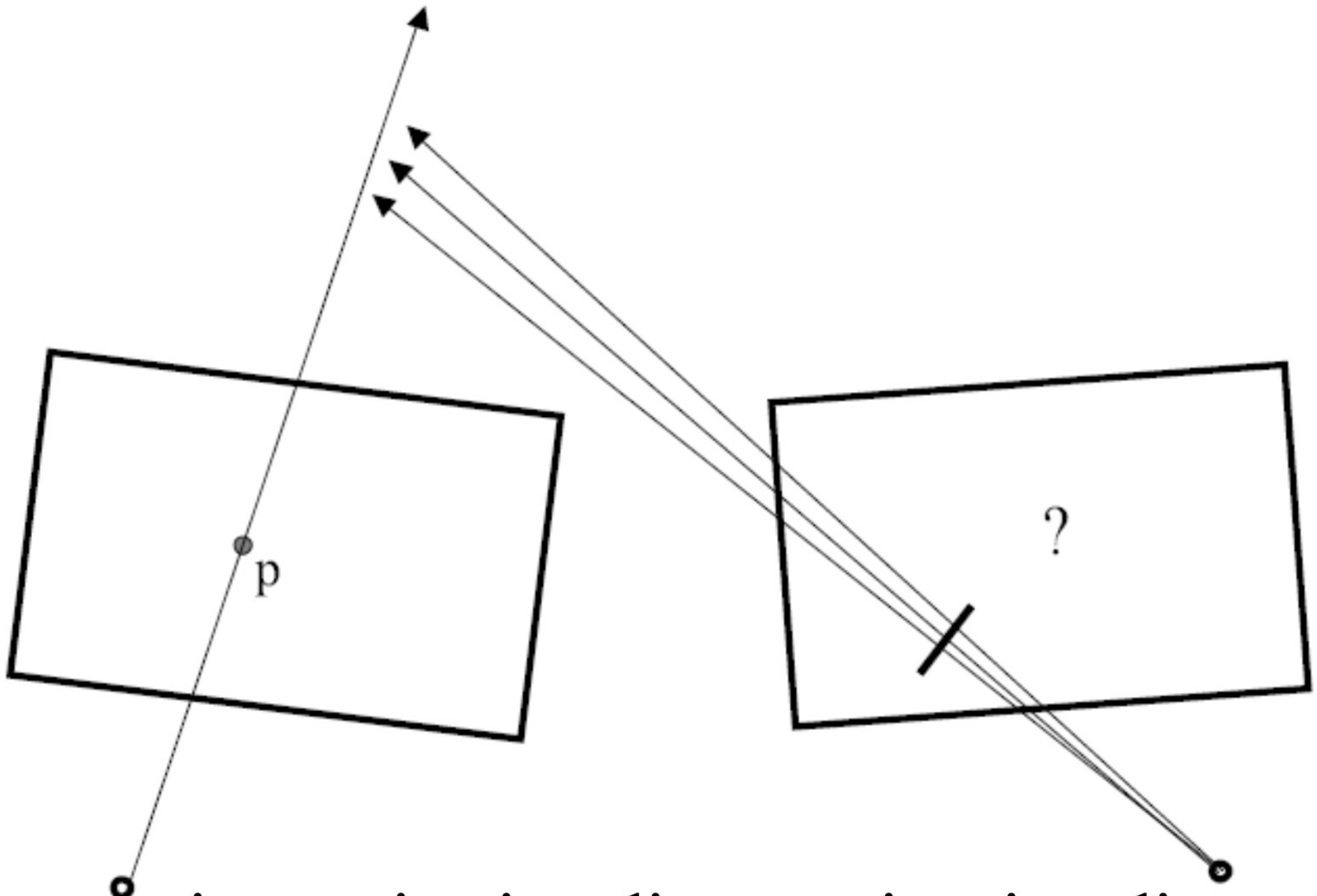
Computer Vision
Structure from Motion


Hier klicken Blick ins Buch!
SECOND EDITION
Multiple View Geometry
in computer vision
Richard Hartley and Andrew Zisserman

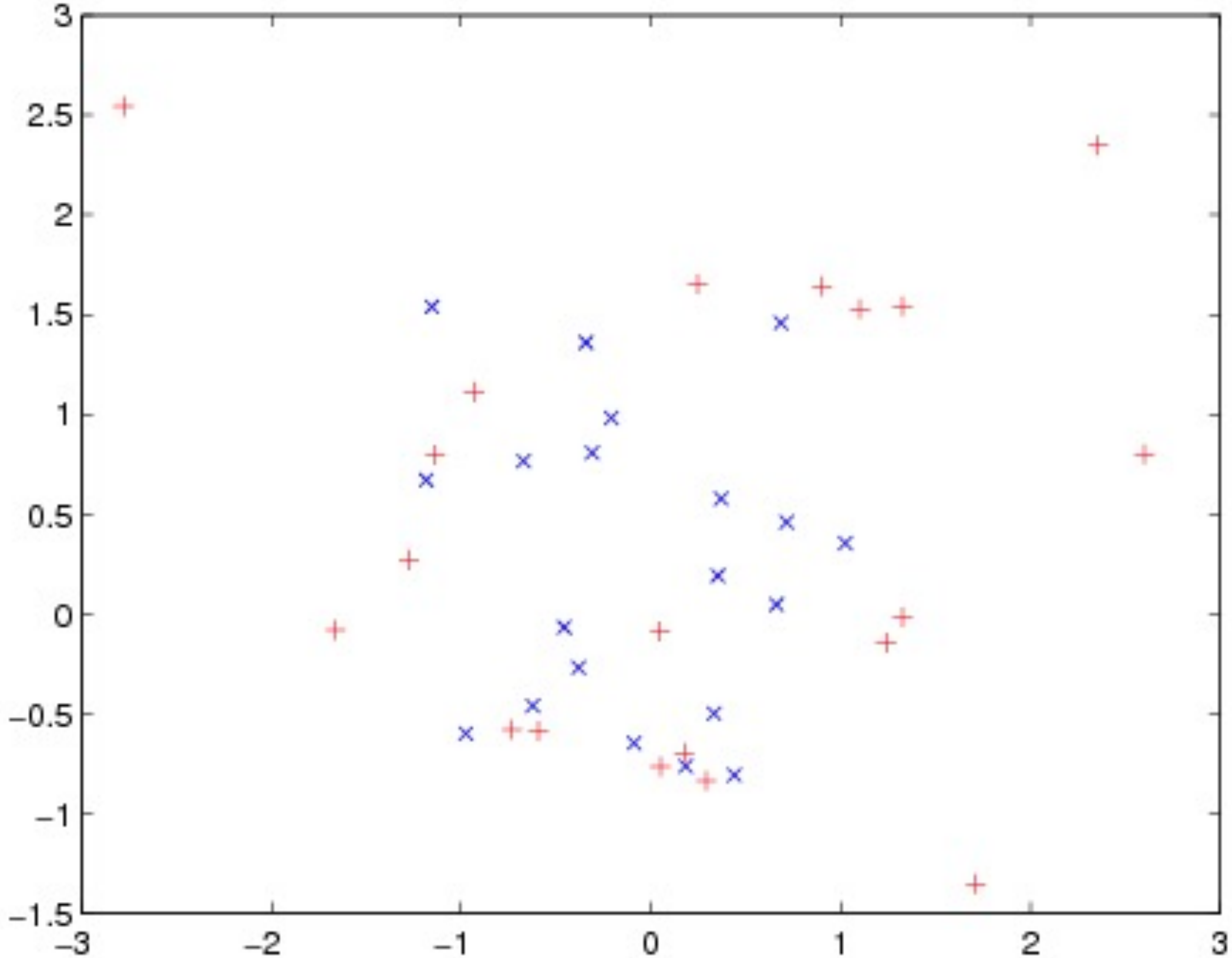

INTERDISCIPLINARY APPLIED MATHEMATICS
IMAGING, VISION, AND GRAPHICS
An Invitation to 3-D Vision
From Images to Geometric Models
Yi Ma
Stefano Soatto
Jana Kosecka
Shankar S. Sastry
Springer

- Structure and depth are inherently ambiguous from single views.





- In perspective projection, lines project into lines. So the line containing the center of projection and the point p in the left image must project to a ***line*** in the right image.


3
2.5
2
1.5
1
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0
−0.5
−1
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−3
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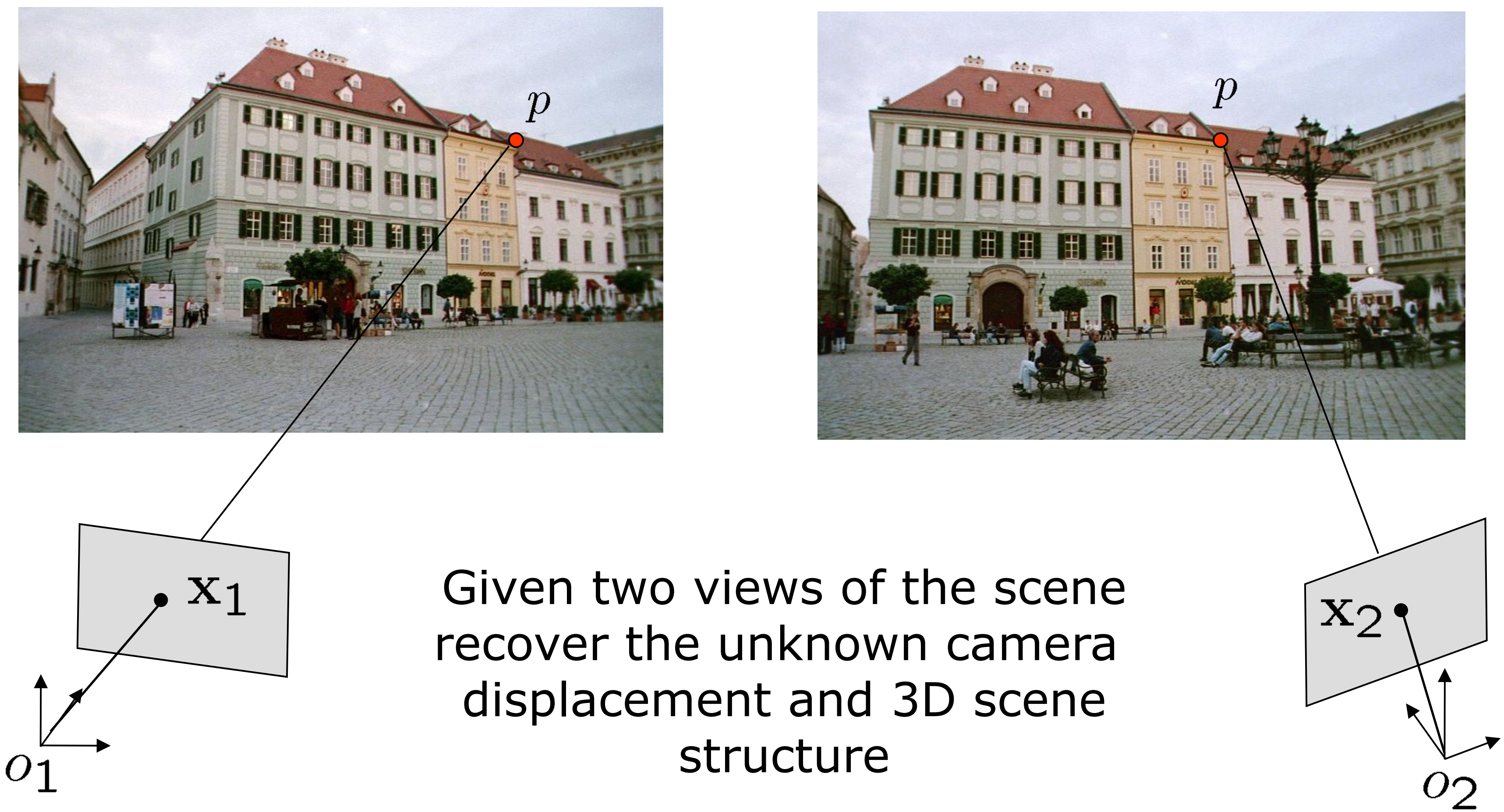
0
1
2
3

General Formulation



Given two views of the scene recover the unknown camera displacement and 3D scene structure

Pinhole Camera Model

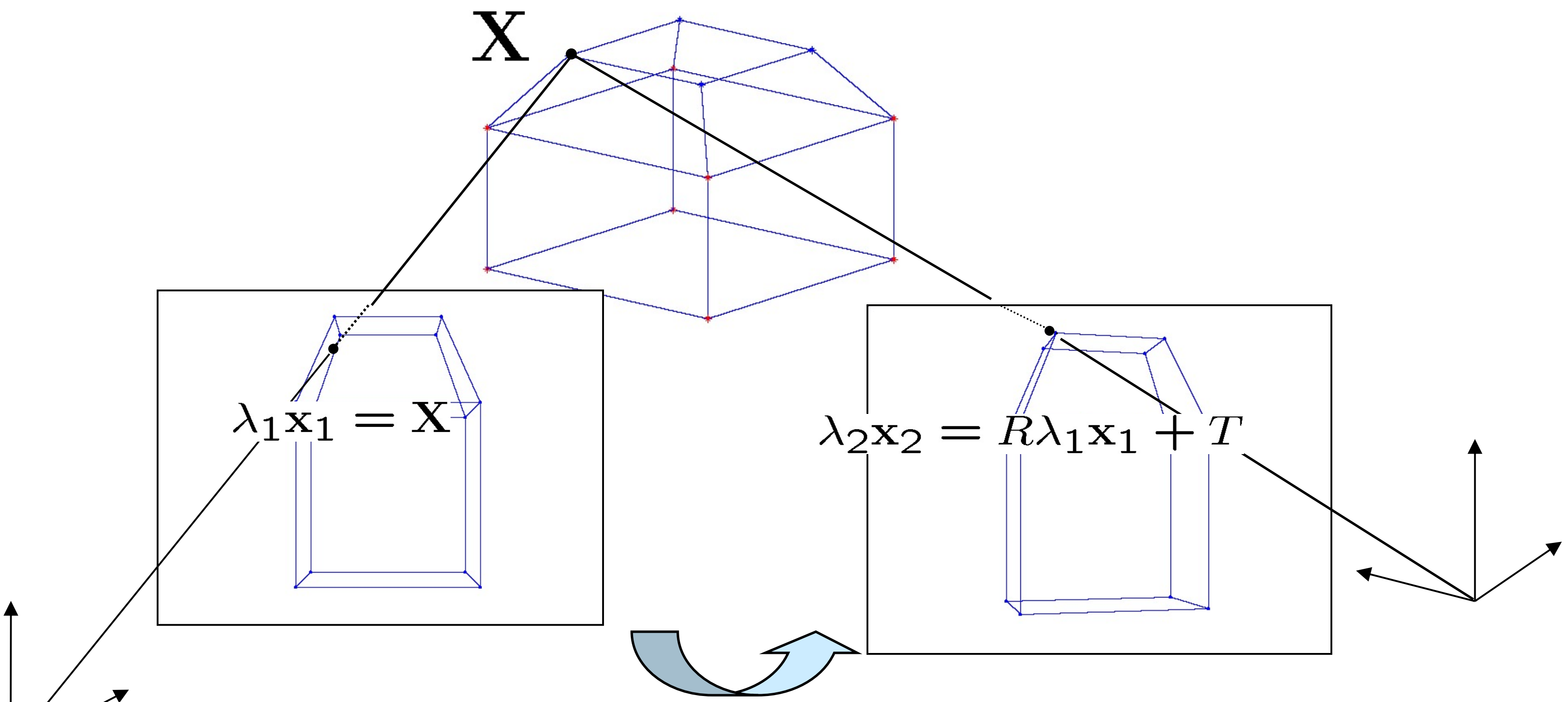
- 3D points $\mathbf{X} = [X, Y, Z, W]^T \in \Re^4, \quad (W = 1)$
- Image points $\mathbf{x} = [x, y, z]^T \in \Re^3, \quad (z = 1)$
- Perspective Projection $\lambda \mathbf{x} = \mathbf{X}$

$$\lambda = Z \;\; x = \frac{X}{Z} \;\; y = \frac{Y}{Z}$$

- Rigid Body Motion $\Pi = [R, T] \in \Re^{3 \times 4}$
- Rigid Body Motion + Projective projection

$$\lambda \mathbf{x} = \Pi \mathbf{X} = [R, T]\mathbf{X}$$

Rigid Body Motion – Two views



$$\boxed{\lambda_2 \mathbf{x}_2 = R\lambda_1 \mathbf{x}_1 + T}$$

3D Structure and Motion Recovery

Euclidean transformation

$$\lambda_2 \mathbf{x}_2 = R \lambda_1 \mathbf{x}_1 + T$$

measurements unknowns

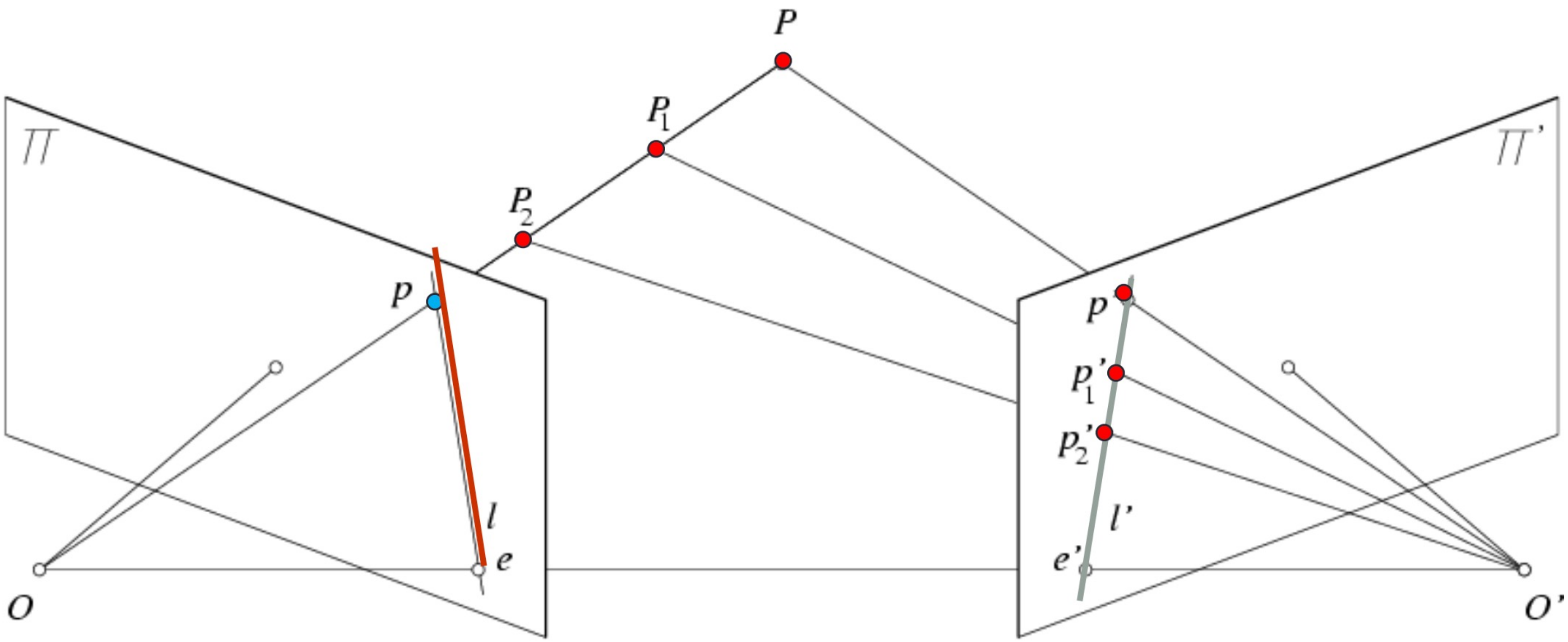
$$\sum_{j=1}^{n} \|\mathbf{x}_1^j - \pi(R_1, T_1, \mathbf{X})\|^2 + \|\mathbf{x}_2^j - \pi(R_2, T_2, \mathbf{X})\|^2$$

Find such Rotation and Translation and Depth that the reprojection error is minimized

Two views ~ 200 points
6 unknowns – Motion 3 Rotation, 3 Translation
- Structure 200x3 coordinates
- (-) universal scale

Difficult optimization problem

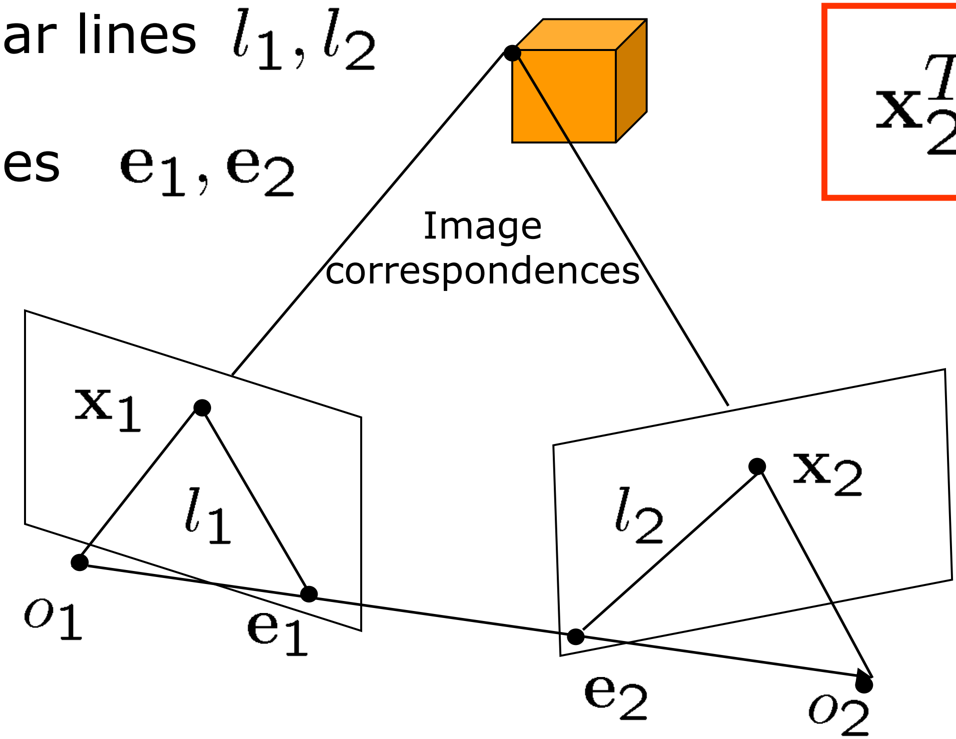
Epipolar constraint



Geometry of two views constrains where the corresponding pixel for some image point in the first view must occur in the second view.

Epipolar Geometry

- Epipolar lines l_1, l_2
- Epipoles $\mathbf{e}_1, \mathbf{e}_2$



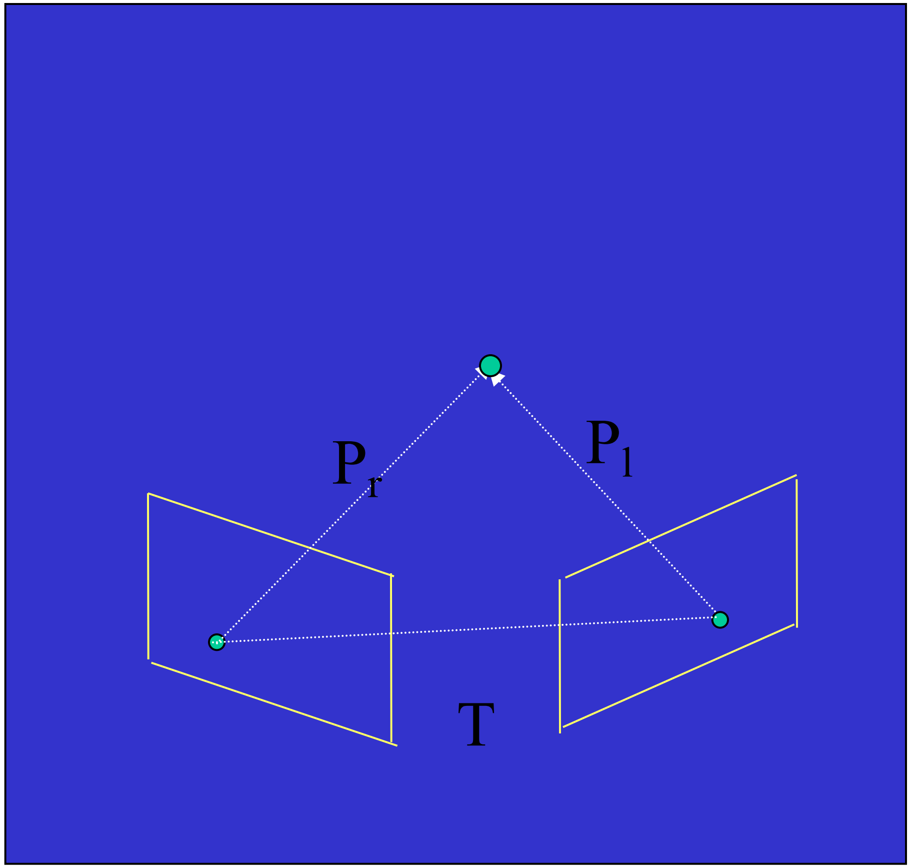
$$\boxed{\mathbf{x}_2^T E \mathbf{x}_1 = 0}$$

$$E = \hat{T} R$$

$$l_1 \sim E^T \mathbf{x}_2 \qquad l_i^T \mathbf{x}_i = 0 \qquad l_2 \sim E \mathbf{x}_1$$

$$E \mathbf{e}_1 = 0 \qquad l_i^T \mathbf{e}_i = 0 \qquad \mathbf{e}_2 E^T = 0$$

EPIPOLAR GEOMETRY: DERIVATION



$P_r = R(P_l - T)$

$$(P_l - T)^T \cdot T \times P_l = 0$$
$$P_r^t R\, T \times P_l = 0$$
$$P_r^t E\, P_l = 0$$

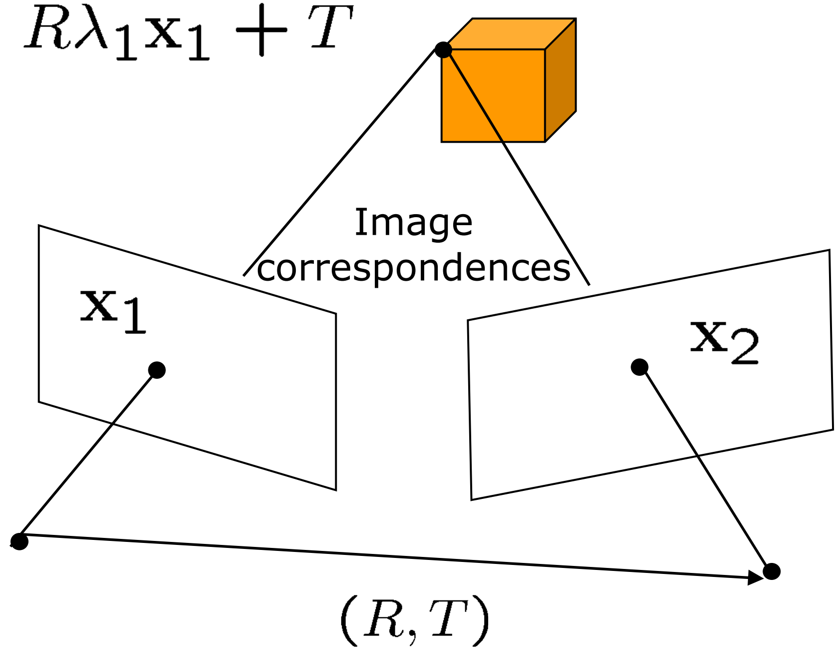
where $E = R\, sk(T)$

$$sk(T) = \begin{pmatrix} 0 & -T_z & T_y \\ T_z & 0 & -T_x \\ -T_y & T_x & 0 \end{pmatrix}$$

The matrix E is called the *essential* matrix and completely describes the epipolar geometry of the stereo pair

Epipolar Geometry

$$\lambda_2 \mathbf{x}_2 = R \lambda_1 \mathbf{x}_1 + T$$



- Algebraic Elimination of Depth [Longuet-Higgins '81]:

$$\mathbf{x}_2^T \underbrace{\widehat{T} R}_{E} \mathbf{x}_1 = 0$$

- Essential matrix $E = \widehat{T} R$

Matrix

$$\mathbf{x}_2^T \hat{T} R \mathbf{x}_1 = 0$$

- Essential matrix $E = \hat{T} R$ Special 3x3 matrix

$$\mathbf{x}_2^T \begin{bmatrix} e_1 & e_2 & e_2 \\ e_4 & e_5 & e_6 \\ e_7 & e_8 & e_9 \end{bmatrix} \mathbf{x}_1 = 0$$

Theorem 1a (Essential Matrix Characterization)
A non-zero matrix E is an essential matrix iff its SVD: $E = U \Sigma V^T$
satisfies: $\Sigma = diag([\sigma_1, \sigma_2, \sigma_3])$ with $\sigma_1 = \sigma_2 \neq 0$ and $\sigma_3 = 0$
and $U, V \in SO(3)$

Estimating the Essential Matrix

- Estimate Essential matrix $E = \hat{T}R$
- Decompose Essential matrix into R, T

$$\mathbf{x}_2^T \hat{T} R \mathbf{x}_1 = 0$$

- Given n pairs of image correspondences:
  - Find such Rotation and Translation that the epipolar error is minimized

$$min_E \sum_{j=1}^{n} \mathbf{x}_2^{jT} E \mathbf{x}_1^j$$

- Space of all Essential Matrices is 5 dimensional
- 3 Degrees of Freedom – Rotation
- 2 Degrees of Freedom – Translation (up to scale !)

Matrix

Essential matrix $E = \hat{T}R$

Theorem 1a (Pose Recovery)
There are two relative poses (R, T) with $T \in \mathbb{R}^3$ and $R \in SO(3)$
corresponding to a non-zero matrix essential matrix.

$$E = U\Sigma V^T$$

$$(\hat{T}_1, R_1) = \quad (UR_Z(+\tfrac{\pi}{2})\Sigma U^T, UR_Z^T(+\tfrac{\pi}{2})V^T)$$
$$(\hat{T}_2, R_2) = \quad (UR_Z(-\tfrac{\pi}{2})\Sigma U^T, UR_Z^T(-\tfrac{\pi}{2})V^T)$$

$$\Sigma = diag([1,1,0]) \quad R_z(+\tfrac{\pi}{2}) = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

- Twisted pair ambiguity $(R_2, T_2) = (e^{\hat{u}\pi}R_1, -T_1)$

Estimating Essential Matrix

$$\mathbf{x}_2^T \widehat{T} R \mathbf{x}_1 = 0$$

- Denote $\mathbf{a} = \mathbf{x}_1 \otimes \mathbf{x}_2$

$$\mathbf{a} = [x_1x_2, x_1y_2, x_1z_2, y_1x_2, y_1y_2, y_1z_2, z_1x_2, z_1y_2, z_1z_2]^T$$

$$E^s = [e_1, e_4, e_7, e_2, e_5, e_8, e_3, e_6, e_9]^T$$

- Rewrite $\mathbf{a}^T E^s = 0$

- Collect constraints from all points

$$\chi E^s = 0$$

$$min_E \sum_{j=1}^{n} \mathbf{x}_2^{jT} E \mathbf{x}_1^j \Rightarrow min_{E^s} \|\chi E^s\|^2$$

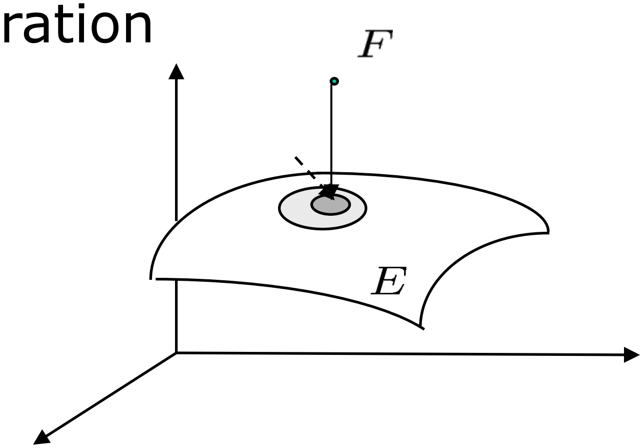
Estimating Essential Matrix

$$min_E \sum_{j=1}^{n} (\mathbf{x}_2^{jT} E \mathbf{x}_1^j)^2 \Rightarrow min_{E^s} \|\chi E^s\|^2$$

Solution

- Eigenvector associated with the smallest eigenvalue of $\chi^T \chi$
- if $rank(\chi^T \chi) < 8$ degenerate configuration

Projection on to Essential Space



Theorem 2a (Project to Essential Manifold)
If the SVD of a matrix $F \in \mathcal{R}^{3\times 3}$ is given by $F = Udiag(\sigma_1, \sigma_2, \sigma_3)V^T$ then the essential matrix E which minimizes the Frobenius distance $\|E - F\|_f^2$ is given by $E = Udiag(\sigma, \sigma, 0)V^T$ with $\sigma = \frac{\sigma_1 + \sigma_2}{2}$

Quadratic Surface

A second-order algebraic surface given by the general equation

$$a x^2 + b y^2 + c z^2 + 2 f y z + 2 g z x + 2 h x y + 2 p x + 2 q y + 2 r z + d = 0. \quad (1)$$

Quadratic surfaces are also called quadrics, and there are 17 standard-form types. A quadratic surface intersects every plane in a (proper or degenerate) conic section. In addition, the cone consisting of all tangents from a fixed point to a quadratic surface cuts every plane in a conic section, and the points of contact of this cone with the surface form a conic section (Hilbert and Cohn-Vossen 1999, p. 12).

Examples of quadratic surfaces include the cone, cylinder, ellipsoid, elliptic cone, elliptic cylinder, elliptic hyperboloid, elliptic paraboloid, hyperbolic cylinder, hyperbolic paraboloid, paraboloid, sphere, and spheroid.

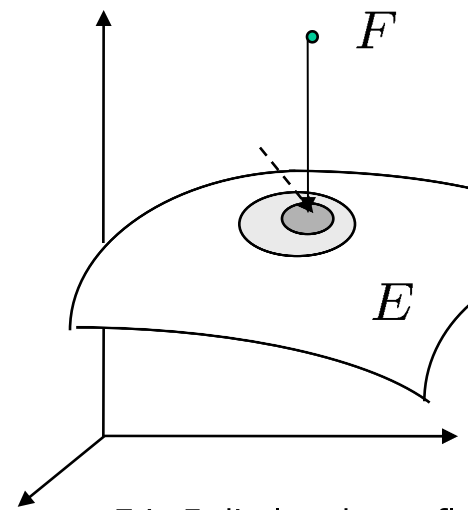
Two view linear algorithm

$$E = \{\hat{T}R | R \in SO(2), T \in S^2\}$$

- Solve the LLSE problem:

$$min_E \sum_{j=1}^{n} \mathbf{x}_2^{jT} E \mathbf{x}_1^j$$

$\chi E^s = 0$ followed by projection



E is 5 diml. sub. mnfld. in

- Project onto the essential manifold:

SVD: $F = U\Sigma V^T$

$$\Sigma' = diag(1,1,0)$$

$$E = U\Sigma' V^T$$

- 8-point linear algorithm

- Recover the unknown pose:

$$\boxed{(\hat{T}, R) = (UR_Z(\pm\tfrac{\pi}{2})\Sigma U^T, UR_Z^T(\pm\tfrac{\pi}{2})V^T)}$$

3D structure recovery

$$\lambda_2 \mathbf{x}_2 = R\lambda_1 \mathbf{x}_1 + \gamma T$$

- Eliminate one of the scale's

$$\lambda_1^j \widehat{\mathbf{x}_2^j} R \mathbf{x}_1^j + \gamma \widehat{\mathbf{x}_2^j} T = 0, \qquad j = 1, 2, \ldots, n$$

- Solve LLSE problem

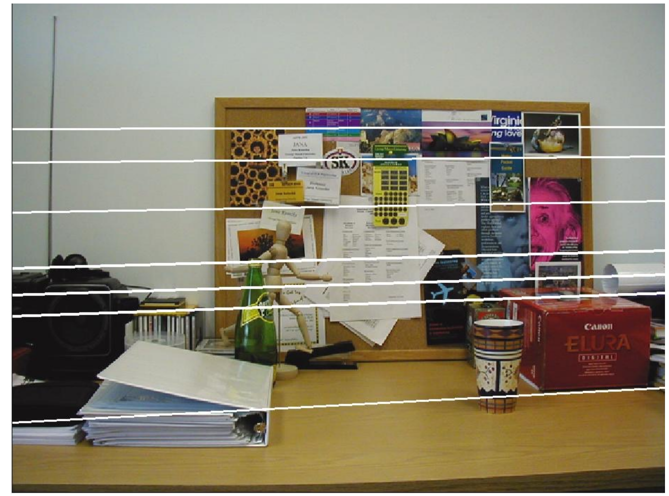
$$M^j \bar{\lambda}^j \doteq \left[\widehat{\mathbf{x}_2^j} R \mathbf{x}_1^j, \ \widehat{\mathbf{x}_2^j} T\right] \begin{bmatrix} \lambda_1^j \\ \gamma \end{bmatrix} = 0$$

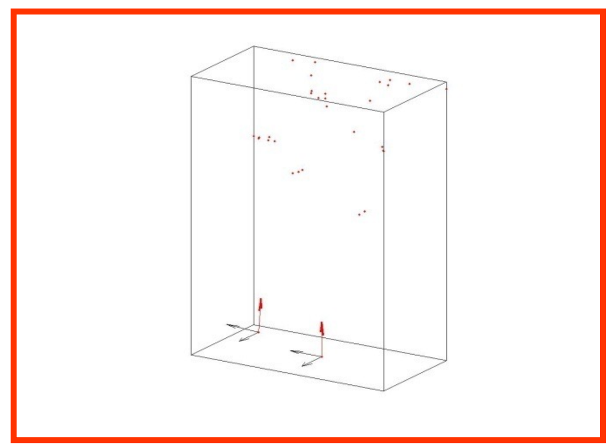
If the configuration is non-critical, the Euclidean structure of then points and motion of the camera can be reconstructed up to a universal scale.

Example- Two views

Point Feature Matching

Example – Epipolar Geometry





Camera Pose
and
Sparse Structure Recovery

Epipolar Geometry – Planar Case

- Plane in first camera coordinate frame

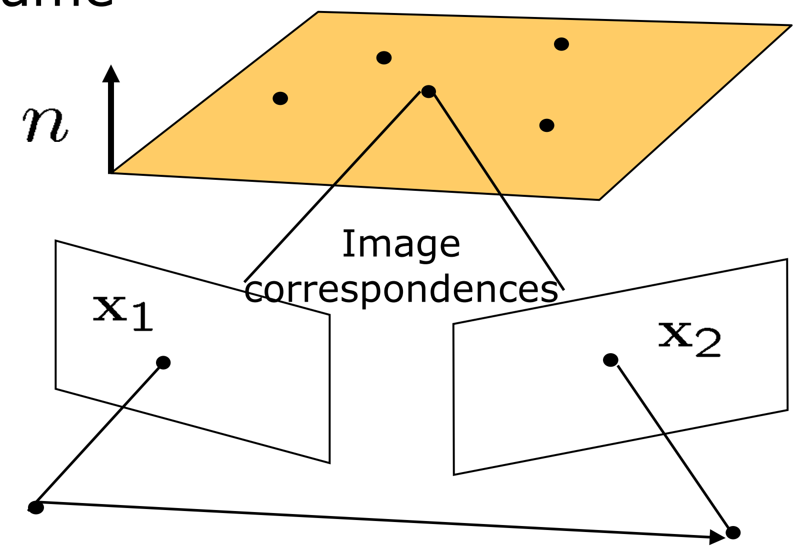
$$aX + bY + cZ + d = 0$$

$$\frac{1}{d} N^T \mathbf{X} = 1$$

$$\lambda_2 \mathbf{x}_2 = R \lambda_1 \mathbf{x}_1 + T$$

$$\lambda_2 \mathbf{x}_2 = (R + \frac{1}{d} T N^T) \lambda_1 \mathbf{x}_1$$

$$\mathbf{x}_2 \sim H \mathbf{x}_1$$



Planar homography

$$\boxed{H = (R + \frac{1}{d} T N^T)}$$

Linear mapping relating two corresponding planar points in two views

Decomposition of H

- Algebraic elimination of depth $\widehat{\mathbf{x}_2} H \mathbf{x}_1 = 0$
- H_L can be estimated linearly $H_L = \lambda H$
- Normalization of $H = H_L / \sigma_3$
- Decomposition of H into 4 solutions $H = (R + \frac{1}{d} T N^T)$

$R_1 = W_1 U_1^T$	$R_3 = R_1$	$R_2 = W_2 U_2^T$	$R_4 = R_2$
$N_1 = \widehat{v_2} u_1$	$N_3 = -N_1$	$N_2 = \widehat{v_2} u_2$	$N_4 = -N_2$
$\frac{1}{d} T_1 = (H - R_1) N_1$	$\frac{1}{d} T_3 = -\frac{1}{d} T_1$	$\frac{1}{d} T_2 = (H - R_2) N_2$	$\frac{1}{d} T_4 = -\frac{1}{d} T_2$

- $$H^T H = V \Sigma V^T \quad V = [v_1, v_2, v_3] \quad \Sigma = diag(\sigma_1^2, \sigma_2^2, \sigma_3^2)$$
$$u_1 \doteq \frac{\sqrt{1-\sigma_3^2} v_1 + \sqrt{\sigma_1^2 - 1} v_3}{\sqrt{\sigma_1^2 - \sigma_3^2}} \quad u_2 \doteq \frac{\sqrt{1-\sigma_3^2} v_1 - \sqrt{\sigma_1^2 - 1} v_3}{\sqrt{\sigma_1^2 - \sigma_3^2}}$$
$$U_1 = [v_2, u_1, \hat{v}_2 u_1], \quad W_1 = [Hv_2, Hu_1, Hv_2 Hu_1];$$
$$U_2 = [v_2, u_2, \widehat{v_2} u_2], \quad W_2 = [Hv_2, Hu_2, \widehat{Hv_2} Hu_2].$$

Motion and pose recovery for planar scene

- Given at least 4 point correspondences $\widehat{\mathbf{x}_2^j} H \mathbf{x}_1^j = 0$
- Compute an approximation of the homography matrix H_l^s
- As nullspace of χ

$\chi H_l^s = 0$ the rows of χ are $\mathbf{a} = \mathbf{x}_1^j \otimes \widehat{\mathbf{x}_2^j}$

- Normalize the homography matrix

$$H = H_L / \sigma_3$$

- Decompose the homography matrix

$$H^T H = V \Sigma V^T$$

- Select two physically possible solutions imposing positive depth constraint